Exp 6 Error correction at Data link layer

Aim: write a program to implement error detection and correction using Hamming code concept

Error correction at Data Link layer:

Hamming code is a set of error-correcting code that can be used to detect and correct errors that can occur when the data is transmitted from the sender to receiver. It is technique developed by R. W. Hamming for error correction.

Program:

```
def encode(data):
    m = len(data)
    r = 0
    while (1 << r) < m + r + 1:
        r += 1
    n = m + r
    code = [0] * (n+1)
```

```
    j = 0
    for i in range(1, n+1):
        if i & (i-1):
            code[i] = data[j]
            j += 1
    for p in [1<<i for i in range(r)]:
        code[i] = data[j]
        j += 1
    for p in [1<<i for i in range(r)]:
        code[p] = sum(cod[i] for i in range(1,n+1)
    return code[1:]

def correct(recv):
    n = len(recv)
    bits = [0] + recv[:]
    Syndrome = 0
    p = 1
    while p<=n:
        if (sum(bits[i] for i in rang(1,n+1)
                if i & p) %2:
            Syndrome += p
        p <<=1
```

```
    if 0 < syndrome <= n:
        bits[syndrome] ^= 1
    return bits[1:], syndrome

if __name__ == '__main__':
    data = [int(b) for b in input("Data bits
    code = encode(data)
    print("Encoded:", "".join(map(str,
    err = int(input(f"Error position bit
    (0-{len(code)}): ") or 0)
    recv = code[:]
    if 0 < err <= len(recv):
        recv[err-1] ^= 1
    fixed, syn = correct(recv)
    print("Received:", "".join(map(str,
    print("Syndrome:", syn)
    print("Corrected:", "".join(map(str,
    print("Data:", "".join(map(str,[b for
        b in extract
```

I/O

Data bits : 1011

Encoded : 011011

Error position to inject (0-7) : 5

Received : 0110111

Syndrome : 5

corrected : 0110011

Data : 1011

Result :

Code Executed Successfully